

Insert new Annex 176A through Annex 176D corresponding to Clause 176 as follows:

Annex 176A

(informative)

SM-PMA test vectors

This annex provides test vectors produced by the transmit function (defined in 176.4.2) of the 200GBASE-R 8:1, 400GBASE-R 16:2, 800GBASE-R 32:4, and 1.6TBASE-R 16:8 SM-PMAs.

Files containing the test vectors are available at <https://standards.ieee.org/downloads/802.3/>.

Editor's note (to be removed prior to publication):

The test vectors are given as plain-text files, which can be found at the following link:

https://www.ieee802.org/3/dj/public/24_06/loewenthal_3dj_02_2406.zip.

This file will be placed in the 802.3 downloads folder referenced above when this amendment is published.

Editor's note (to be removed prior to publication):

The format and content of the test vectors is described in the presentation at the following link:

https://www.ieee802.org/3/dj/public/24_06/loewenthal_3dj_01a_2406.pdf

In the subsequent draft, this annex will be updated with content describing the test vectors.

The test vectors are provided at vector points VP1, VP2, VP3 shown in Figure 176A–1, for the case when there is a PCS or XS sublayer directly above the PMA.

The test vectors at VP1 are for the m PCS lanes at the input of the $m:n$ SM-PMA.

Test vectors at VP2 are after the odd PCS lanes have gone through the delay function defined in 176.4.2.4. VP2 test vectors are not present for the 1.6TBASE-R 16:8 PMA since it does not implement the delay function.

Test vectors at VP3 are for the n lanes at the output of the symbol-multiplexer defined in 176.4.2.5.

The following assumptions are made when generating the test vectors:

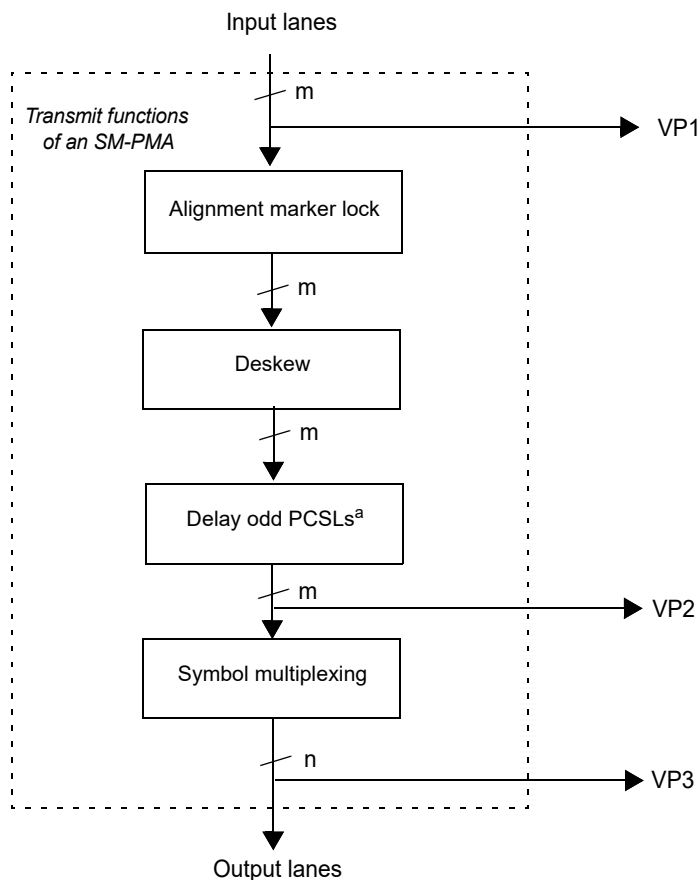
- The test vectors contain a total of four RS-FEC codewords worth of data, which is 21 760 bits, distributed across the lanes at each of the vector points.
- The first of the four codewords contains the alignment marker group.
- The test vector data that is not from an alignment marker consists of Idle control characters
- The skew between PCS lanes at VP1 is assumed to be zero.
- The seed of the PRBS9 generator used for the alignment marker padding bits is $P<0:8> = 0x100$.
- The scrambler seed for the 200 and 400 Gb/s PMAs is set to $S<0:57> = 0x24E6959D0FA5DBD$
- The scrambler seed for the 800 Gb/s and the 1.6 Tb/s PMA flow 0 is set to $S0<0:57> = 0x24E6959D0FA5DBD$, and for flow 1 it is set to $S1<0:57> = 0x1FB58857D816246$.
- $tx_am_sf<0:2>$ in the AM block is set to $\{0, 0, 0\}$ for all the test vectors.

For the 200 Gb/s, 400 Gb/s and 800 Gb/s m:n SM-PMAs, 8 PCS lanes are symbol-multiplexed into an output PMA lane n using the following order:

- 200 Gb/s 8:1 PMA: PCS lanes 0, 1,...7.
- 400 Gb/s 16:2 PMA: PCS lanes $n \times 8$, $n \times 8 + 1$, $n \times 8 + 2$, ..., $n \times 8 + 7$, where $n = 0, 1$.
- 800 Gb/s 32:4 PMA: PCS lanes $n \times 4$, $n \times 4 + 16$, $n \times 4 + 1$, $n \times 4 + 16 + 1$, ..., $n \times 4 + 3$, $n \times 4 + 16 + 3$, where $n = 0, 1, 2, 3$.

For the 1.6 Tb/s 16:8 SM-PMA, 2 PCS lanes are multiplexed into an output PMA lane n in the order $n \times 2$, $n \times 2 + 1$, where $n = 0, 1, 2, \dots, 7$.

The test vectors at VP1 and VP2 are provided in a text file per PMA output lane. Each line in the files is 40 bits in binary format, and belongs to a PCS lane that is multiplexed into the PMA output lane. The order of the PCS lanes in the VP1 and VP2 test vector files is the same as the PCS lane multiplexing order described above for the four SM-PMAs. The VP3 test vector file is also per PMA lane. Each line contains 160 bits. In the VP2 and VP3 test vector files, the character X is used to denote “don’t care” and represents bits on odd numbered PCS lanes that are present as a result of adding delay to the odd numbered PCS lanes.



^a If 200GBASE-R, 400GBASE-R or 800GBASE-R

$m = 8$ (200GBASE-R), 16 (400GBASE-R), 32 (800GBASE-R), or 16 (1.6TBASE-R)
 $n = 1$ (200GBASE-R), 2 (400GBASE-R), 4 (800GBASE-R), or 8 (1.6TBASE-R)

Figure 176A–1—Test vector points for the 200GBASE-R 8:1, 400GBASE-R 16:2, 800GBASE-R 32:4, and 1.6TBASE-R 16:8 PMAs